Multidimensional Goodhart

Response Channels and Residual Shape

Kerkko 'Xylix' Pelttari

Abstract. Goodhart's law is often summarized as proxy failure under optimization. The motivating hypothesis here was more specific: in multidimensional proxy systems, optimizing a set of measured dimensions should make the remaining error directions more structured and perhaps more complicated. Working through selection, intervention, and aggregation models narrows that hypothesis. The surviving object is not a universal complexity law, but response-dependent error geometry. Pure selection gives baseline response curves and chi-square/value drift bounds; intervention replaces baseline statistics with action, cost, search, and stakes primitives; additive and conjunctive scorecards produce different exchange-rate and entry effects. The useful transfer rule is therefore a response-modeling contract, checked against a literature map that treats external Goodhart, ML, and incentive results as primitive-specific analogues rather than generic support.

1 Introduction: the original hypothesis

The familiar slogan is that when a measure becomes a target, it ceases to be a good measure [6, 16]. The original research hypothesis behind this paper was a multidimensional mathematical version of that slogan: when a proxy system optimizes some set of measured dimensions, the error in the remaining unmeasured or undermeasured dimensions should not merely grow. It should acquire structure. As successive proxy dimensions are modelled, measured, or optimized away, residual error should migrate into the other dimensions in a more complicated geometric pattern.

In informal terms, the expectation was recursive Goodhart with geometry. A scalar proxy leaves a residual. Adding dimensions constrains some visible failure modes. But optimization pressure may then move into the residual degrees of freedom that the expanded proxy still does not control. If this is right, a multidimensional model should reveal how error becomes redistributed, concentrated, rerouted, or made less legible as the measured coordinate set changes.

That broad hypothesis is too elastic as stated. More dimensions do not by themselves imply more failure. Selection over a fixed baseline is not the same mechanism as agents changing behavior at fixed type. A compensatory scorecard does not have the same geometry as a conjunctive one. And "more complicated" is not a prediction until the response process and the complexity or shape measure are fixed in advance. Scalar tail-conditioned Goodhart already has sharp external treatments, notably El-Mhamdi and Hoang's top-α selection analysis under target/discrepancy assumptions [4]; the contribution here is a multidimensional response-modeling envelope and transfer rule.

The narrower result of the modelling work is therefore this: residual error has a response-dependent geometry. The core object is not bare dimension count, but the channel that connects optimization pressure to movement in state space. Selection, intervention, and aggregation each impose different mathematical constraints on the shape of residual error. The most reusable pieces are the response-channel split, the selection and value-drift bounds, the separation between action-cost budgets and hidden welfare, the scorecard exchange-rate condition, and the response-modeling contract that says which primitive must be declared before a calculation transfers.

Write the target-relevant state as $G(s) \in \mathbb{R}^m$, the proxy as $P(s) \in \mathbb{R}^k$, the intended proxy relation as $\varphi : \mathbb{R}^m \to \mathbb{R}^k$, and the residual as $\varepsilon(s) = P(s) - \varphi(G(s))$. A proxy can fail through hidden goal directions in $\ker \varphi$, through proxy residuals, or through agents moving after the score is used for control. The framework below separates those cases by the response channel.

> The original hypothesis was that multidimensional Goodhart pressure would make off-proxy error more complicated as measured dimensions are optimized. The licensed result is sharper and more conditional: the shape of residual error is governed by the declared response channel, action/search geometry, aggregation rule, and hidden value model. The framework does not imply that more metrics are generically harmful or helpful; it asks which response routes the measurement system opens, closes, weights, or makes cheap.

2 Relation to existing formulations

The literature review changes how the contribution should be stated. Goodhart's monetary-policy warning, Campbell's social-indicator warning, and Strathern's target/measure slogan are genealogy and motivation, not theorem statements [1, 6, 16]. Manheim and Garrabrant's taxonomy separates regressional, extremal, causal, and adversarial variants [12], but a taxonomy does not by itself supply a response channel, cost model, aggregation rule, or welfare functional.

The scalar mathematical anchor is El-Mhamdi and Hoang's top-α selection analysis under a target/discrepancy split [4]. Their theorems give sharp tail-conditioned asymptotics under independence and named tail families. The selection bound below is different: it is a looser chi-square envelope for arbitrary baseline dependence and declared hidden coordinates. Majka and El-Mhamdi's independence-free scalar formalization is the closest external neighbor to that independence stance, but it remains scalar and discrepancy-structured [11]. Smith and Winkler's optimizer's curse is a precursor selection-bias inequality, not an intervention or welfare model [15].

The late-stage formal-analogue map gives the same discipline for ML and economics results. Lucas's critique and performative prediction support the idea that policy can change the data-generating response channel [10, 13]. Strategic classification gives an action-cost analogue of fixed-type response [7]. Holmstrom–Milgrom multitask incentives are the closest economics-side precedent for measured-task substitution and aggregation geometry [9]. Skalse et al.'s reward-hacking result is a proxy/target primitive, not a welfare or action-geometry theorem [14]. None of these sources is used below as generic authority; each maps to a primitive and carries a non-transfer boundary.

3 Selection channels

In a pure selection channel, the policy reweights a fixed baseline law. Let $(S, \mathcal{F}, \mu)$ be the baseline space, $H : S \to \mathbb{R}^d$ hidden coordinates, and $P : S \to \mathbb{R}$ the scalar proxy. A selection policy has nonnegative weight W_θ with finite positive expectation, likelihood ratio $L_\theta = \frac{W_\theta}{\mathbb{E}_{\mu[W_\theta]}}$, selected law $\mu_\theta = L_\theta \mu$, and hidden response

$$B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu[H]}.$$

Hard thresholds use $W_t = \mathbb{1}\{P \geq t\}$ and define the threshold response $b_{H(t)} = \mathbb{E}[H \mid P \geq t] - \mathbb{E}[H]$. Boltzmann selection uses $W_\beta = \exp(\beta P)$ only on the finite-mgf domain $\mathcal{B} = \left\{\beta : \mathbb{E}_{\mu[\exp(\beta P)]} < \infty\right\}$. Where differentiation is valid,

$$\mathrm{d}\mathbb{E}_{\beta[H]} / \mathrm{d}\beta = \mathrm{Cov}_{\beta(H,P)},$$

so covariance is a local velocity along a selection path, not a finite-pressure primitive. The toy example $P = Z$, $H = Z^2 - 1$ has zero baseline covariance but nonzero threshold and finite Boltzmann response.

The Gaussian threshold model gives the cleanest coordinate picture. If $X \sim \mathcal{N}(0, \Sigma)$, $P = X_1$, and $H = (X_2, \ldots, X_m)$, then

$$\mathbb{E}[H \mid X_1 \geq t] = \sigma_1 \lambda(t/\sigma_1) r,$$

where $r_j = \frac{\mathrm{Cov}(X_j, X_1)}{\mathrm{Var}(X_1)}$. Dimensional growth appears only if the coupling-vector norm grows with the number of hidden coordinates; a bounded total coupling budget blocks it.

Selection bounds. If $\delta = \|L_\theta - 1\|_{L^2(\mu)} = \sqrt{\chi^2(\mu_\theta \parallel \mu)}$ and hidden coordinate standard deviations are s_i, then

$$\left|B_{H_i}(\theta)\right| \leq \delta s_i, \quad \left\|B_{H(\theta)}\right\|_2 \leq \delta \|s\|_2.$$

For a declared value vector v,

$$|\Delta(v \cdot H)| \leq \delta \sqrt{v^T \Sigma_H v}.$$

The first form is coordinate-explicit Euclidean drift. The second is declared value/operator drift; it is the right form when the scalar welfare direction is part of the model.

These are baseline-only claims. They stop at reweighting. They do not analyze what fixed agents can do after a metric is announced.

4 Intervention channels

Intervention begins when the policy changes behavior at fixed type. Declare a type space U, baseline type law ν, baseline kernel $K_0(ds \mid u)$, participation weights $W_{\theta(u)}$, and policy-indexed response kernels $K_{\theta(ds|u)}$. The induced law is

$$\mu_{\theta(A)} = \left(\int W_{\theta(u)} K_{\theta(A|u)} \nu(du)\right) / \left(\int W_{\theta(u)} \nu(du)\right).$$

Pure selection relative to (U, K_0) has $K_\theta = K_0$ almost everywhere and all policy dependence in W_θ. Intervention has $K_\theta \neq K_0$ on a positive-mass set of types. Mutual singularity with baseline is sufficient evidence that baseline-only selection fails, but it is not the definition: absolute continuity can survive when the baseline already contains epsilon mass on the response image.

The smallest algebraic intervention toy is a Stackelberg threshold game. An agent has true quality Q, can choose action $a \geq 0$, receives score $Q + a$, passes if $Q + a \geq t$, gains prize V, and pays cost $\frac{a^2}{2\kappa}$. In the noiseless case, agents with $Q \in [t - \Delta, t)$ game just enough to pass, with

$$\Delta = \sqrt{2\kappa V}.$$

The wedge is controlled by ease of gaming and stakes, not by baseline covariance. It is a toy signature of quadratic threshold gaming, not a law of neural training or RLHF.

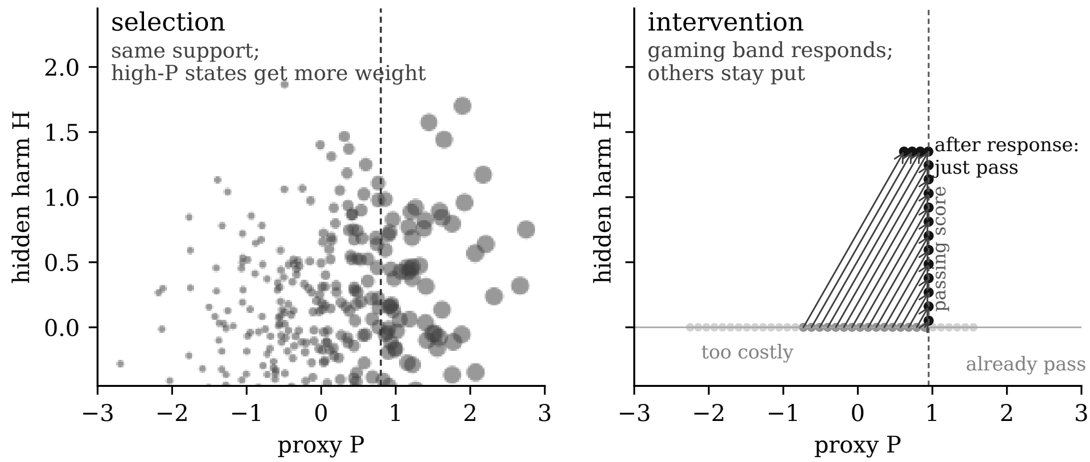

Figure 1: Selection reweights fixed support; intervention transports fixed types into new states. The distinction is why the selection-channel drift bound has no baseline-only analogue for intervention channels.

For a more general local intervention budget, declare an action a, convex cost $c(a)$, linear score gain $w \cdot a$, and score deficit d. The score-deficit cost is

$$m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)],$$

under the usual finite-dimensional convex regularity. Gaming the deficit is privately feasible when $m(d) \leq V$. This is not a welfare bound until a hidden harm functional has also been declared.

5 Multidimensional scorecards

Adding measured dimensions has no sign by itself. In an additive scorecard with measured set M, separable quadratic costs $\sum_{j \in M} \frac{a_j^2}{2\kappa_j}$, score $\sum_{j \in M} w_j a_j$, and hidden harm $\sum_{j \in M} h_j a_j$, the cost-minimal action for fixed deficit d is

$$a_j^* = d\kappa_j w_j / W_M, \quad W_M = \sum_{i \in M} \kappa_i w_i^2,$$

and fixed-deficit per-agent harm is

$$H_{\mathrm{per}(M,d)} = d \cdot \left(\sum_{j \in M} h_j \kappa_j w_j \right) / \left(\sum_{j \in M} \kappa_j w_j^2 \right).$$

Thus $H_{\mathrm{per}(M,d)}$ is conserved across active measured sets if and only if $h_j = c w_j$ on the channels being compared. In the unit-weight equal-harm case, re-routing conserves fixed-deficit per-gamer harm, but increasing the aggregate gaming capacity can expand entry and raise population harm $H_{\mathrm{pop}(M,F_Q,V)}$.

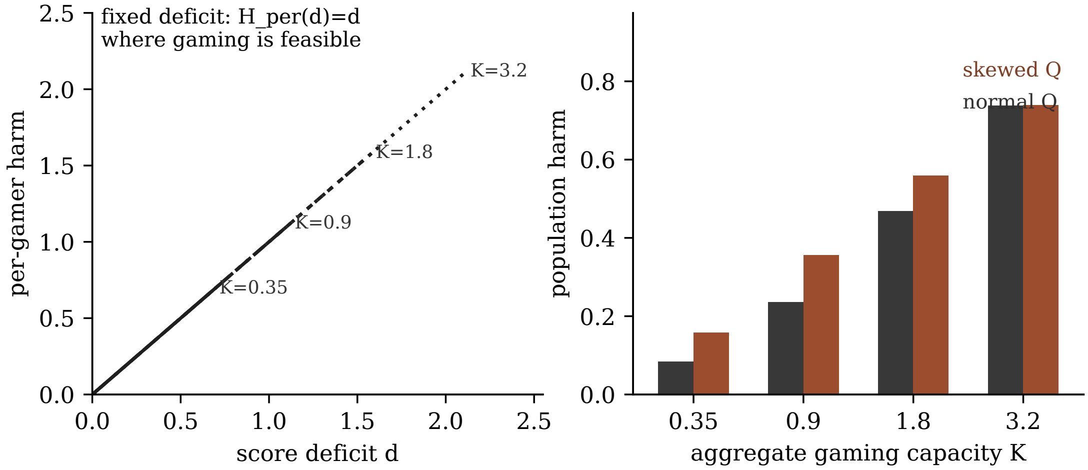

Figure 2: In the unit-weight equal-harm additive model, feasible fixed-deficit $H_{\mathrm{per}(d)}$ curves coincide, while H_{pop} grows as gaming capacity expands the profitable deficit band.

A conjunctive scorecard flips the per-gamer comparison. If passing requires $a_j \geq t$ for every $j \in M$, then $H_{\mathrm{per}}^{\mathrm{conj}(M,t)} = t|M|$ under equal harm. The population sign still depends on entry, because the cost of passing also rises. The contrast is the point: aggregation and exchange rates determine the distortion geometry.

6 Response-modeling contract and MMLU

The surviving framework is methodological. A claim should declare: type space, baseline behavior, policy exposure, response channel, action/search geometry, proxy and target, aggregation rule, hidden harm, and evidence standard. This contract is the transfer rule for applications. Domains do not inherit toy bounds by analogy; they inherit the obligation to declare the primitives that make a calculation or falsifier meaningful.

MMLU is a useful worked example because one score can support several different channels [8]. Selecting among fixed checkpoints is a selection claim over candidates and may use W_θ plus declared hidden value weights. Finetuning, prompt search, contamination, synthetic data filtering, and reward/proxy optimization are response-kernel claims and require K_θ plus action/search geometry. Model-selection overfitting [2], adaptive holdout reuse [3], strategic classification [7], performative prediction [13], and reward-model overoptimization [5] are nearby formal warnings, but none by itself supplies κ, V, a convex cost, or a welfare functional.

> MMLU is Layer-2 application discipline, not empirical proof. It shows how the contract separates fixed-checkpoint selection from finetuning/search/ contamination/reward-proxy mechanisms. It does not license Stackelberg, convex-cost, RLHF, or welfare bounds without additional primitives and falsifiers.

7 Open problems

Several problems remain deliberately open, while one formerly open adaptive hardening subcase is now only a boundary note.

- **Welfare-bound packaging.** Convex private affordability and realized cost-minimizing harm are distinct from worst-case welfare bounds for a declared hidden-harm functional.
- **Endogenous value and stakes.** V may depend on observed gaming, trust, institutional response, or future proxy repair. The exogenous-V Stackelberg and convex-cost calculations do not solve that dynamic problem.
- **Adaptive hardening.** In the finite-channel deterministic additive-score toy with fixed M, fixed deficit d, fixed stakes V, fixed weights, separable quadratic costs, deterministic obser-

vation, and monotone progress, fixed-deficit gaming stops exactly when $S_{t(M)} = \sum_j \kappa_{j,t} w_j^2 < d^2/(2V)$. Changing measured sets or deficits, stochastic observation, endogenous stakes, shared bottlenecks, nonconvex geometry, and policy optimization remain separate problems.
- **Empirical implementation.** Applications need repeated-type observations, randomized or staggered exposure, action traces, cost variation, or pre-registered hidden outcomes.
- **Non-convex ML and RLHF mappings.** Gradient accessibility, benchmark contamination, reward-model simplicity, and optimizer search are not interchangeable substitutes for κ or convex costs.

Iterations 20–21.1 provide deterministic Layer-3 checks for the toy regimes: linear-Gaussian and nonlinear selection response, finite-mgf Boltzmann behavior, declared value vectors, the Stackelberg wedge, quadratic water-filling, noisy response diagnostics, welfare-bound separation, active-face quadratic repairs, capped fixed-charge switches, and additive/conjunctive population checks. They are toy checks of the framework's bookkeeping, not empirical confirmation of a general Goodhart law.

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